Night and Day: Diurnal Warming and Structural Transformation in India

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Abstract

This paper finds diverging partial effects of diurnal warming – higher nighttime (T_{min}) and daytime (T_{max}) temperatures – on the share of agricultural wage-labor in the Indian Census (1981–2011). Decadally, both warmer nights and days contract cultivated area and grain output; only warmer days raise harvest prices locally. Warmer nights consolidate agriculture, shifting seasonal workers and self-cultivators into wage labor; warmer days casualize it, pushing wage labor back to seasonal work. Long differences show the labor divergence is significant only in rural populations. In towns, warmer nights and days depress nonagricultural worker shares.

Keywords: Climate Change, Structural Transformation, Agriculture, India
JEL Codes: O1, Q54, J43, R11

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The reallocation of labor from agriculture into non-agricultural work—structural transformation—is the foundation of economic development (Matsuyama 1992; Herrendorf, Rogerson, and Valentinyi 2014). Whether a warming climate arrests it is unsettled, and the evidence splits by horizon. In the short run, heat pushes workers off the field: Colmer (2021) finds that year-to-year heat shocks move Indian labor out of agriculture within the year, and in rural Mexico Jessoe, Manning, and Taylor (2017) finds that extreme heat depresses farm and local non-farm employment alike, pushing workers to migrate out. Over the long run the sign flips—reading six decades of the Census, Liu, Shamdasani, and Taraz (2023) finds that sustained warming in India holds labor on the land and depresses non-farm demand, while across countries Nath (2025) shows that when heat lowers farm productivity a "food problem" locks economies into agricultural specialization. Within India, then, the two horizons disagree on the central margin of development.

Yet the escape valve that clears the shock in Mexico is largely shut in India: rural–urban mobility is strikingly low, as households forgo large urban wage gains to keep the caste-based insurance networks that out-migration would sever (Munshi and Rosenzweig 2016). Displaced labor cannot simply leave; it must be absorbed locally. And the stakes are first-order—agriculture still employs close to half of India's workforce, so the direction in which heat moves that labor decides whether warming hastens India's structural transformation or stalls it.

Prior literature measures heat with a daily average, a count of degree days, or a tally of extreme days (Schlenker, Hanemann, and C. 2006; Blakeslee and Fishman 2015; Dell, Jones, and Olken 2014), which sum two partial effects that need not share a sign. The nighttime minimum and daytime maximum have themselves warmed at different rates in India: since the early 1990s the former has risen faster than the latter, narrowing the diurnal range across several agro-climatic zones, including the Indo-Gangetic crop belt (Mall et al. 2021). Crops stand in the field around the clock; the farmers that tend to them work only in daylight. A warmer night raises nocturnal respiration and the overnight heat load the crop carries, spending after dark the carbon it fixes by day (Peng et al. 2004; Welch et al. 2010); a warmer day presses field labor against the physiological ceiling of sustained outdoor exertion, lowering the productivity of labor itself (Dunne, Stouffer, and John 2013; Somanathan et al. 2021). These are dry-bulb, air-temperature channels, and the responses estimated here are decadal adaptations rather than single-season yield shocks: over a decade cultivation adjusts to the warming it faces, so the estimates recover the net direction of labor reallocation, not the sign of any one season's yield. This paper holds the two margins apart, estimating the partial effect of the nighttime minimum (T_{min}) and the daytime maximum (T_{max}), each net of the other and of rainfall, across Indian Census districts, towns, and urban households from 1981 to 2011. A higher minimum compresses the diurnal range and a higher maximum widens it; where the daily average collapses the cycle to a single mean, the decomposition recovers its range—the quantity agronomy ties to grain filling and heat stress.

I establish five central empirical facts; the general-equilibrium framework of Section 1 is built to rely on them, its assumptions dictated by the data rather than imposed on it.

First, the composition of agricultural employment reverses between night and day. In a decadal district panel, a 1°C rise in the nighttime minimum raises the agricultural wage-labor share by 4.4 percentage points, drawing workers from self-cultivation (-2.8) and the seasonal margin (-3.3); an equal rise in the daytime maximum reverses each sign, cutting the wage-labor share by 2.6 points and returning 3.2 to the seasonal pool. The two coefficients reject equality ($F = 24.9$ for wage labor, 14.7 for seasonal), and the reversal is the diurnal cycle's own: a single growing-season degree-days measure collapses it to one muted coefficient ($+0.036$). It survives a two-decade long difference ($+0.075/-0.062$) and humidity-adjusted wet-bulb temperatures ($+0.086/-0.079$).

Second, the two margins move the farm economy asymmetrically. Both contract cultivated area by indistinguishable amounts (-0.22 and -0.22, $F = 0.01$), while only the daytime maximum raises the local harvest price ($+0.13$) and depresses the field wage (-0.12), the nighttime minimum leaving both flat. Daytime heat acts as a classic supply shock—scarcer output, dearer grain, cheaper labor—while nighttime warming rearranges who works the land at an unchanged price.

Third, the reversal scales with the weight of agriculture in local employment: sharp among rural workers ($F = 26.3$) and muted among urban ($F = 6.0$), the same sign in both, the amplitude tracking the agricultural share of the workforce.

Fourth, the divergence closes in towns. Over a long difference, both margins depress the town non-agricultural share—by 3.8 points under warmer nights and 7.8 under hotter days—so each end of the cycle holds labor out of non-agriculture.

Fifth, suggestive evidence points to where the displaced labor lands: matched to urban households in three rounds of the National Sample Survey, it arrives at the informal, casual end of urban work—petty trade, transport, construction—rather than in manufacturing or salaried employment (Appendix Table A11).

What unites these facts is that nighttime and daytime warming reach the farm through different factors of production. A warmer night acts on land, scaling the productive services of the soil; a warmer day lowers total factor productivity, eroding the capacity for sustained physical work. When labor and land are gross substitutes—their elasticity of substitution above one—the two shocks reach labor demand through different channels: a land-augmenting shock alters the land-to-labor ratio and shifts the demand for agricultural labor directly, independent of its effect on yields, whereas a Hicks-neutral productivity loss leaves that ratio intact and contracts only the scale of production. The two thus move agricultural labor demand in opposite directions—warmer nights raise it, drawing self-cultivators and seasonal workers into agricultural wage labor, and warmer days lower it, returning laborers to the seasonal margin. Under unit elasticity, the Cobb–Douglas case, the two shocks are observationally equivalent and the divergence cannot arise; gross substitutability is the economic content of the result.

This leaves a natural alternative to rule out: that the divergence reflects market structure rather than factor bias, with rural districts behaving as closed subsistence economies and urban districts as open ones. A general-equilibrium treatment reverses that intuition. In a closed economy the subsistence demand for food makes its price rise more than one-for-one as output contracts, and the resulting income effect draws labor into agriculture under daytime heat and releases it under nighttime warming—the opposite of the observed orientation. The signs in the data can arise only when the local food market is integrated enough for trade to set the price, so the orientation itself places India's rural and urban districts alike in the open regime, and it is partial integration that lets a daytime output loss raise the local harvest price—why only the daytime margin moves it. Here the framework departs from the closed-economy "food problem" of Liu, Shamdasani, and Taraz (2023) and Nath (2025): under sustained heat the food market is open, so the reallocation runs through factor bias and the seasonal margin rather than a subsistence price spiral. Its amplitude then scales with the agricultural share of local employment and the depth of the seasonal margin—large among rural workers, muted among urban. The framework fixes the direction of each response; the magnitudes are the data's to supply.

This paper makes four contributions. First, the diurnal decomposition reconciles the conflicting short- and long-run estimates of Colmer (2021) and Liu, Shamdasani, and Taraz (2023): warming sorts agricultural labor rather than pushing it in a single direction, and which way it moves depends on which end of the day warms. Second, a general-equilibrium framework with an explicit openness parameter shows the night/day divergence to be a property of factor bias—preserved in open economies, reversed in closed ones—separating it from the closed-economy "food problem" (Liu, Shamdasani, and Taraz 2023; Nath 2025) and locating India's districts, by the orientation of their estimated signs, in the open regime. Third, I construct a consistent-boundary panel of Indian towns from the digitized Census and link it to urban household surveys for suggestive evidence on the informal destinations of displaced agricultural labor that district aggregates leave unresolved. Fourth, the paper qualifies the structural-transformation literature (Matsuyama 1992; Herrendorf, Rogerson, and Valentinyi 2014): under sustained heat, towns foreclose entry into formal non-agricultural employment while agriculture intensifies and specializes—a reallocation of labor that stalls, rather than advances, the structural transition. Section 1 develops the framework; the next section describes the data and the decomposition; Section 3 reports the district, town, and household results; and the last concludes.

1. Conceptual Framework

This section lays out the economic logic that maps each temperature margin to the labor responses the rest of the paper estimates. Appendix A formalizes it and Table A1 collects the predicted signs.

What the two margins measure. The nighttime minimum and the daytime maximum are the floor and the ceiling of the daily temperature cycle. Holding one fixed, a higher minimum compresses the diurnal range and a higher maximum widens it, so the decomposition speaks to the range—the quantity agronomy ties to grain filling and heat stress—as much as to the level. The two margins act on different inputs. Crops occupy the field around the clock, so a warmer night works on the standing crop and the land in cultivation, raising respiration and the overnight heat load the crop carries (Peng et al. 2004; Welch et al. 2010). Field labor occupies only the daylight, so a warmer day works on the worker, pressing effort against the physiological ceiling of sustained outdoor exertion and lowering the productivity of labor itself (Dunne, Stouffer, and John 2013; Somanathan et al. 2021). A daily average folds the two channels into one number; the analysis keeps them apart.

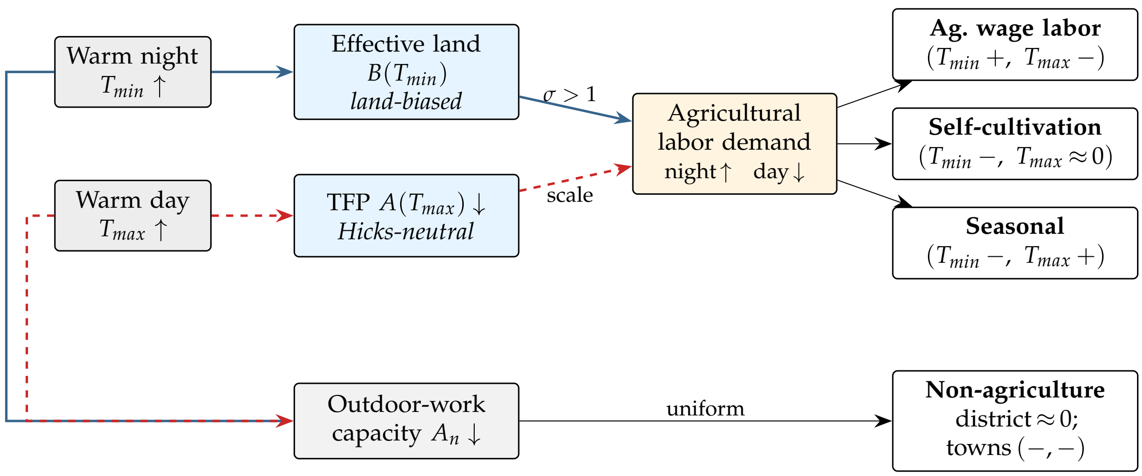
Two shocks, two factors. The framework assigns each margin to the factor it acts on: nighttime warming is a *land-biased* shock that raises or lowers the productive services of cultivated land, while daytime warming is a *Hicks-neutral* loss of total factor productivity. The distinction has bite only because labor and land are gross substitutes—the farm can trade one for the other, an elasticity of substitution σ above one. When a warmer night changes the quality of an acre, a farm that substitutes readily reworks its land-to-labor mix and works the land harder, hiring *more* field labor rather than less; the direction of the labor response is set by the factor the shock lands on, not by what happens to yields. A neutral productivity loss lands on no factor in particular: it leaves the land-to-labor ratio intact and shrinks the scale of the operation, pulling down the marginal product of labor and the field wage and, where the food market is tight, raising the harvest price. The two margins therefore move farm-labor demand in *opposite* directions—warmer nights raise it, warmer days lower it. That opposition is the entire content of the decomposition, and gross substitution is what makes it possible: under Cobb–Douglas ($\sigma = 1$) a land-augmenting shock folds into total factor productivity and becomes observationally neutral, the two channels collapse, and a daily average loses nothing.

Where the labor goes: the seasonal sink. Workers divide across four states—agricultural wage labor, self-cultivation, non-agricultural work, and casual seasonal work—whose shares sum to one, so the margins move as one offsetting system, and the casual seasonal pool is the flexible residual that clears the reallocation. When nighttime warming raises the demand for farm labor, the farm hires: the wage-labor share rises and draws from the cultivators who give up their marginal plots and the casual workers who attach to the harvest—the signature of a commercializing farm that retains and reorganizes labor rather than releasing it. When daytime warming collapses the marginal product, the wage-labor market gives way and returns workers to the seasonal margin. Self-cultivation falls under both margins, as marginal plots go unworked.

Open and closed economies. Whether this orientation survives depends on the food market the district sits in. In a closed economy the local price clears against local supply, and because households must eat, that price moves more than one-for-one with output; in an open economy the district takes the price set by trade across the wider state. The two regimes disagree on sign, not merely magnitude. A closed subsistence economy in fact *inverts* the orientation: a daytime output loss drives the local food price up so sharply that the income effect pulls labor *into* agriculture, while nighttime warming releases it—the reverse of what the data show. The observed signs therefore require the local food market to be open beyond a threshold degree of integration, and they place India's rural and urban districts alike on the open side of it. Because the night and day shocks reach the price through different channels, the nighttime response is the more demanding test of that openness: it is the night sign, not the day, that pins down how open the economy must be. Partial integration still does the essential work—it is what lets a daytime output loss raise the local harvest price, and so why only the daytime margin moves it.

Rural areas, towns, and the reading of the results. The rural–urban contrast is then one of amplitude, not of sign: the reversal is large where agriculture dominates employment and a deep casual-labor margin absorbs the reallocation, and small where agriculture is marginal and a thick non-farm market cushions the shock. In towns, where farm work is too small to register the reversal, the shared loss of capacity for outdoor informal work shows through instead, as a uniform contraction of non-agricultural employment under both margins—the long-run restraint of non-farm demand under heat (Liu, Shamdasani, and Taraz 2023; Nath 2025). The framework belongs to the structural-transformation tradition of Matsuyama (1992), Gollin and Rogerson (2014), and Bustos, Caprettini, and Ponticelli (2016); Appendix A develops it in full and Table A1 collects the predicted signs, with proofs in the supplemental online appendix. Figure 1 traces each channel to the labor share it moves. These signs guide the analysis rather than serving as a structure to be estimated; the next sections test them.

FIGURE 1. The diurnal mechanism: pathways from warming to each labor margin



Solid blue marks the nighttime (T_{min}) pathway and dashed red the daytime (T_{max}) pathway. A higher T_{min} augments or degrades effective land; because labor and land substitute ($\sigma > 1$), it raises agricultural labor demand and, through the seasonal sink, lifts the wage-labor share while draining self-cultivation and casual work. A higher T_{max} is a neutral productivity loss that shrinks the scale of the operation, lowering labor demand and returning workers to the seasonal margin. Both margins also erode outdoor-work capacity (A_n), which in towns—where farm labor is too small to register the reallocation—shows through as a uniform contraction of non-agriculture; both additionally contract cultivated area (the common supply shock, omitted as it is not a labor share). Signs in each outcome box are the estimated (T_{min}, T_{max}) responses from the district panel.

2. Data and Empirical Approach

2.1. Data Sources

I harmonize four datasets to link climate, agricultural production and structural transformation: (1) daily gridded weather data from the IMD, (2) district agricultural data (ICRISAT), (3) district and town level tables from the Indian Census, and (4) urban household-level employment data from the NSS. These datasets span across 4 decades (Census, IMD, ICRISAT), with a shorter window from the NSS (1987, 1999, 2009).

Indian Meteorological Department. Gridded data were obtained from the Indian Meteorological Department (IMD) on air temperature (1° resolution) and precipitation (0.25° resolution) between 1951-2023. Consistent with the approach in Liu, Shamdasani, and Taraz (2023), daily observations are averaged over growing seasons of the past 10 years. The growing season marks the months of June-February, which captures cropping cycles of rice in the monsoon (kharif) and wheat in the winter (rabi) seasons. The baseline exposure measures are the growing-season means of the daily minimum and maximum surface-air temperatures, T_{min} and T_{max}, which separate nighttime from daytime warming. For comparison I also construct degree days (DDs), a scalar that rescales and sums agriculturally relevant temperatures between 8 and 32 °C (Blakeslee and Fishman 2015; Schlenker, Hanemann, and C. 2006); degree days average over the diurnal cycle and so collapse the two margins into one.

$$DDs = \sum_d D(\mathrm{T}_d) \text{ ; where } \mathrm{T}_d = \frac{\mathrm{T}_{min,d} + \mathrm{T}_{max,d}}{2} \tag{1}$$

$$D(\mathrm{T}) = \begin{cases} 0, & \text{if } \mathrm{T} \leq 8°\mathrm{C} \\ \mathrm{T} - 8, & \text{if } 8°\mathrm{C} < \mathrm{T} \leq 32°\mathrm{C} \\ 24, & \text{if } \mathrm{T} > 32°\mathrm{C} \end{cases} \tag{2}$$

Census of India. The core dataset is sourced from the District Census Handbooks (DCHBs) of India. Specifically, I digitize the Primary Census Abstract (PCA) tables, covering the universe of Indian towns from 1981 to 2011. A Census town meets three criteria: (1) a minimum population of 4,000, (2) at least 75% of the male working population in non-agricultural activity, and (3) a population density above 400 persons per sq. km. I construct a panel of 2,969 towns that appear consistently across all four decades, so that the analysis reflects intensive-margin adjustments rather than the reclassification of urban areas. For comparability over time, occupations are aggregated into four mutually exclusive categories: cultivators (landowners), agricultural laborers (wage workers), non-agricultural workers, and marginal workers. The "marginal" category cannot be assigned cleanly to agriculture or non-agriculture, but it lets me track the contractual *quality* of work.

ICRISAT. The ICRISAT District Level Database (DLD) tracks agricultural inputs and production in India between 1966-present. I harmonize districts to 2011-Census consistent boundaries, constructing an annual panel of cultivated area, production, yields and harvest prices for key grains: rice, wheat, maize, sorghum, barley and millets. I construct the mean of these measures over 10 years preceding each Census round, e.g. over the years 1971-1980 for the 1981 Census. Thus, the results from the temperature regressions compare contemporaneous horizons of 10-year windows with climate data.

National Sample Surveys. While the Census measures the volume of labor reallocation, it does not resolve the destination sectors within non-agriculture under its 1981 definitions. To identify where displaced labor ends up, I use three rounds of the National Sample Survey (NSS) Employment and Unemployment Schedules (1987, 1999, and 2009), harmonized by IPUMS (Ruggles et al. 2024). I group occupations into "outdoor" work (cultivation, agricultural labor, construction, mining) and "indoor" work (manufacturing, services), and restrict the sample to urban households as a benchmark to the Census town-level data. This linkage reveals the destination of heat-displaced labor—casual and seasonal work rather than stable employment in manufacturing or services.

2.2. Empirical Approach

The identification relies on two approaches: a fixed effects panel estimation (3), and a long differences approach (4) described under Dell, Jones, and Olken (2014), Burke and Emerick (2016) and Liu, Shamdasani, and Taraz (2023). Let Y_{irt} be the outcome of interest (e.g., share of workers in agriculture) for district or town i, within state r, observed at time t. $\mathbf{T}_{it}$ denotes a vector of temperature measures, either degree days $\mathbf{T} = \{DDs\}$ or the diurnal surface-air pair $\mathbf{T} = \{T_{min}, T_{max}\}$. All specifications control for total precipitation $\mathbf{P}_{it}$ and fit a state-specific linear time trend.

$$Y_{irt} = \beta \mathbf{T}_{irt} + \phi \mathbf{P}_{irt} + \mu_i + \lambda_t + \gamma_r \cdot t + \varepsilon_{irt} \tag{3}$$

$$\Delta Y_{ir} = \beta \Delta \mathbf{T}_{ir} + \phi \Delta \mathbf{P}_{ir} + \gamma_r + \varepsilon_{ir} \tag{4}$$

For temporal parity, I construct the gridded temperature and rainfall measures as a moving average over the growing seasons of the preceding ten years. Decadal climate is the inverse-distance-squared–weighted average of IMD grid points surrounding the district (within 500 km) or town (within 250 km) centroid. Because the weights fall off as the squared distance, grid points beyond roughly 100 km contribute negligibly.

Equation 3 is a town-decade panel with two-way fixed effects and a state-specific linear trend; I read its coefficients as the *short-run* (decadal) effects of heat on structural transformation. Equation 4 averages the outcome Y and covariates X over two endpoints—1981–91 and 2001–11—and regresses $\Delta Y = Y_2 - Y_1$ on $\Delta X = X_2 - X_1$. This long-differenced estimator recovers the effect of a sustained increase in $\mathbf{T}$ on the outcome trend ΔY, measured against its secular trend $\Delta \bar{Y}$ over the sample. By absorbing unobserved state-level trajectories $(\gamma_r \cdot t)$, I read these coefficients as the *long-run* (multi-decade) adaptation response of labor to sustained heat.

I enter T_{min} and T_{max} simultaneously and read each coefficient as the partial effect of one margin holding the other—and rainfall—constant. I consider several alternative specifications, including spatially corrected standard errors (Conley 1999). Appendix Tables A12–A14 report rejection rates from regressing the key outcomes on a placebo climate regressor—the real climate series randomly permuted across districts, which preserves its decadal persistence and spatial distribution but breaks any true link to the outcome—and I select the preferred panel estimator for equation 3 as the specification that rejects the null in roughly 5% of simulations. Unadjusted and heteroskedasticity-robust standard errors over-reject sharply, whereas clustering by district with state-specific linear trends restores the nominal 5% size.

The simultaneous estimator addresses two identification challenges. First, because T_{min} and T_{max} are highly correlated ($\rho = 0.93$), entering either alone would induce omitted-variable bias; entering them together isolates their partial effects and identifies a narrowing versus widening of the local diurnal range. Second, absorbing state-specific trends accounts for time-varying state policies and compares towns within their local economic geography under similar baseline climate trajectories. Correlograms of the residuals confirm that the spatial correlation of these heat shocks in the key outcomes decays within a 100 km radius.

3. Results

Table 1 decomposes the effects of warming nights and days on the district agricultural economy: Panel A reads down the supply chain for the major cereals—rice, wheat, maize, sorghum, barley, and millets—and Panel B reports the within-district reallocation of labor. The nighttime minimum and the daytime maximum carry opposite signs across the price and labor outcomes, consistent with the theoretical framework.

Both margins shrink the farm. The contraction in cultivated land is statistically

TABLE 1. Panel Estimates of Nighttime vs Daytime Warming in Districts

Panel A. ICRISAT Districts (1981-2011)

	Log Levels (All Grains)				
	Cultivated Area	Production	Harvest Price Index	Field Wage	Yield Index
	(1)	(2)	(3)	(4)	(5)
$\Delta\ T_{min}$	-0.218***	-0.145**	-0.042	-0.038	0.051
	(0.046)	(0.064)	(0.031)	(0.060)	(0.040)
$\Delta\ T_{max}$	-0.225***	-0.221***	0.130***	-0.119*	-0.046
	(0.056)	(0.064)	(0.034)	(0.070)	(0.047)
Observations	1,243	1,243	1,191	1,115	1,211
Dep Var. Mean	5.45	5.77	5.66	2.86	7.14
F-stat ($T_{min} = T_{max}$)	0.01	0.80	13.34	0.74	2.63

Panel B. Census Districts (1981-2011)

	Share of Workers				
	Ag. Labor	Cultivator	Seasonal	Non-Agriculture	Log Workers
	(1)	(2)	(3)	(4)	(5)
$\Delta\ T_{min}$	0.044***	-0.028**	-0.033***	0.017	0.175**
	(0.010)	(0.012)	(0.012)	(0.011)	(0.088)
$\Delta\ T_{max}$	-0.026***	-0.010	0.032**	0.004	-0.017
	(0.009)	(0.013)	(0.013)	(0.013)	(0.083)
Observations	1,187	1,187	1,187	1,187	1,187
Dep Var. Mean	0.18	0.32	0.17	0.32	13.45
F-stat ($T_{min} = T_{max}$)	24.91	0.85	14.66	0.54	2.36
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: Panel A reports log cultivated area, an area-weighted yield index, log production, the log field wage, and a log harvest-price index for the major Indian cereals (rice, wheat, barley, maize, millets, sorghum). Panel B reports district shares of total workers by category and, in the last column, the log number of district workers. T_{min} and T_{max} are the growing-season (June–February) surface-air minimum and maximum, averaged over the preceding decade; all specifications control for total precipitation over the same window. The data form a decadal panel (1981–2011) on 2011 Census-consistent district boundaries, with district and decade two-way fixed effects and a state $\times$ decade linear trend. Standard errors clustered at the district level.

identical across the two margins—a $1°$C rise cuts log area by 0.22 either way ($F = 0.01$)—while production falls comparably rather than identically (-0.15 at night, -0.22 by day, $F = 0.8$). I read the milder nighttime loss as labor intensification at work: the significant nighttime rise in wage labor adds output back through the production identity, cushioning the extensive-area loss, where daytime labor-shedding compounds it. The margins part on the price. The daytime maximum raises the log harvest price by 0.13 while the nighttime minimum holds it flat, a split sharp at $F = 13.3$; the field wage edges down under both (-0.12 by day, -0.04 at night, too close to separate). Daytime warming acts as a supply shock: the daytime loss of land and grain passes straight into a dearer harvest price, where the nighttime price holds flat. Cereal yields move little under either margin, the flat reading the land-augmentation channel anticipates once the gain to standing crops nets against the lost area.

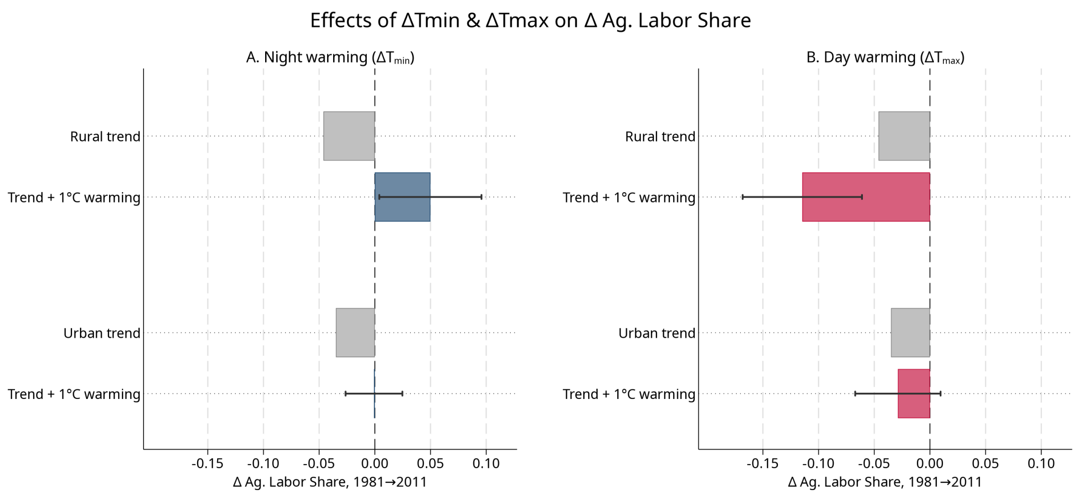
Panel B locates the labor. A warmer night raises the agricultural wage-labor share by 4.4 points and draws that labor off self-cultivation (-2.8) and the casual seasonal margin (-3.3); a warmer day reverses the flow, cutting the wage-labor share by 2.6 points and returning 3.2 points to the seasonal margin. The two margins separate sharply—$F = 24.9$ on wage labor, $F = 14.7$ on the seasonal share—while non-agricultural employment holds flat under both ($F = 0.5$): the district reshuffles labor within agriculture. The extensive count rises at night, a warmer minimum lifting the log number of district workers by 0.18. The appendix framework grounds the split: nighttime warming augments land, and because labor and land are gross substitutes it raises the demand for field labor directly; daytime warming is a neutral productivity loss that shrinks the operation and sheds labor to the seasonal pool.

Higher nighttimes commercialize agriculture. Though the loss in grain production is comparable to higher daytimes, the district heads off the price shock through adaptation: consolidating land and intensifying agricultural labor. Higher daytimes casualize it. Facing a comparable loss, the farm contracts instead of adapting: the wage-labor market gives way, workers fall back to the casual seasonal margin, and the supply loss passes straight through to the harvest-price spike.

Figure 2 carries the labor decomposition to its two-decade horizon and splits each district into its rural and urban populations. The long difference averages each district's two Census endpoints—1981 and 1991 against 2001 and 2011—and differences them, stripping decade-specific noise to isolate the change in climate against the change in labor. The grey bars set the stage: the agricultural wage-labor share fell across rural and urban India over these two decades, the secular structural transformation drawing labor off the land.

Warming bends that trend, and the two margins bend it in opposite directions. A warmer night raises the rural agricultural wage-labor share by 0.096 and turns its -0.046

FIGURE 2. Long-differenced estimates of warming on the declining agricultural wage-labor trend in districts



Notes: Long-difference estimates (1981–2011) of a 1°C rise in growing-season nighttime (T_{min}) and daytime (T_{max}) surface-air temperature on the district agricultural wage-labor share, estimated separately for the rural and urban populations of each district. Grey bars give the secular trend (the mean change in the share); each colored bar adds the warming coefficient to that trend ($\bar{y} + \hat{\beta}$), with 95% confidence intervals. A colored bar whose interval clears its grey trend bar is a statistically significant warming effect. All specifications control for the long-run change in precipitation and include state fixed effects; standard errors are clustered at the district level.

secular decline into a $+0.050$ rise: nighttime warming hoards labor on the farm against the structural trend. A warmer day lowers the same share by 0.068, deepening the decline to -0.114: daytime heat pushes a rural agricultural exit. The opposition is sharp and holds over the two-decade horizon.

The urban populations of the same districts barely move. A warmer night lifts the urban agricultural share by 0.034—a third of the rural response—and a warmer day leaves it flat ($+0.006$, insignificant). The reversal is rural; the town blocks it. The framework reads this as amplitude (Proposition A4): the diurnal response scales with the agricultural share of local employment, deep in rural districts and thin in urban ones, so the same factor-bias force registers sharply in rural areas and faintly in the town. Table A8 reports the full rural and urban decomposition behind the figure—all four worker shares and log total workers. District populations move in the same direction as workers, with the urban population rising under nighttime warming (Table A9).

Table 2 brings the long difference to the Census town. In the balanced town panel (Panel A), where agriculture is a thin share of employment, the diurnal divergence collapses. Both margins now raise the town agricultural wage-labor share—$+0.063$ at night and $+0.041$ by day—and the night/day split falls to $F = 1.6$, against the sharp rural divergence of Figure 2. Too little agriculture remains in the town to register the factor-bias reallocation, and cultivators, seasonal workers, and the worker count hold roughly flat.

The town's signal sits instead on the non-agricultural margin, where both warming measures depress employment (-0.038 at night, -0.078 by day). This is the heat-capacity drag the framework assigns to informal urban work: a productivity loss on outdoor and physical labor that falls under both ends of the cycle, contracting non-agricultural employment without a day/night split. Where the rural district reshuffles labor within agriculture, the town sheds it from the informal non-farm sector—a reallocation that stalls, rather than advances, the structural transition.

Panel B narrows to statutory towns—those under municipal governance as of 1981, India's established urban core. The non-agricultural drag survives and sharpens (-0.061 at night, -0.087 by day, both significant), yet the extensive margin parts ways with Panel A: total workers now *rise* under both margins ($+0.44$, $+0.38$) rather than holding flat. The established town does not shed labor on net—it absorbs it, growing its workforce even as the non-farm *share* contracts. That combination is the footprint of in-migration into towns with deeper labor markets, an adaptation-through-mobility channel the framework brackets—it holds the town workforce fixed—rather than generates.

The appendix confirms the decomposition under alternative measures (Appendix Section C): a single degree-days measure collapses the opposition into one muted coefficient (Table A4); and wet-bulb temperatures sharpen the labor split while humid

TABLE 2. Long-Difference Estimates of Warming on Town Labor Reallocation

	Δ Share of Workers				
	Ag. Labor	Cultivator	Seasonal	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
Panel A: Balanced Census Towns					
$\Delta\ T_{min}$	0.063***	-0.009	-0.016	-0.038**	0.272
	(0.012)	(0.011)	(0.012)	(0.018)	(0.169)
$\Delta\ T_{max}$	0.041***	0.022	0.017	-0.078***	0.241
	(0.016)	(0.014)	(0.014)	(0.022)	(0.171)
Observations	2,969	2,969	2,969	2,969	2,969
Dep Var. Mean	-0.05	-0.06	0.10	0.01	0.55
F-stat ($T_{min} = T_{max}$)	1.61	3.38	4.06	2.78	0.11
Panel B: Statutory Towns (Municipal in 1981)					
$\Delta\ T_{min}$	0.053***	-0.001	0.005	-0.061***	0.436**
	(0.012)	(0.013)	(0.012)	(0.019)	(0.187)
$\Delta\ T_{max}$	0.031*	0.022	0.035**	-0.087***	0.377**
	(0.018)	(0.016)	(0.017)	(0.024)	(0.188)
Observations	2,159	2,159	2,159	2,159	2,159
Dep Var. Mean	-0.05	-0.06	0.10	0.01	0.56
F-stat ($T_{min} = T_{max}$)	1.45	1.78	2.74	1.02	0.35
Rainfall Control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: The table reports long differences (1981–2011) of each Census town worker share on the long-run change in T_{min} and T_{max}, in place of the decadal panel. Panel A is the balanced set of Census towns (urban in every census 1981–2011); Panel B restricts to statutory towns—those holding municipal status as of the 1981 census—the established urban core. The four mutually exclusive Census categories—agricultural wage labor, cultivators, seasonal, and non-agricultural—are exhaustive and sum to one, so the share coefficients sum to approximately zero. Column 5, set apart by a gap, reports the change in log total workers (the extensive margin). The F-statistic tests $T_{min} = T_{max}$ for each outcome. T_{min} and T_{max} are the growing-season (June–February) surface-air minimum and maximum; all specifications control for the long-run change in precipitation and include state fixed effects. Standard errors clustered at the district level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

nights raise cereal yields outright ($+0.15$), the land-augmenting signature ($B' > 0$) that the flat dry-bulb yield only implies (Table A5).

4. Conclusion

This paper asks which way warming pushes structural transformation. I decompose it into nighttime and daytime surface-air components and read the labor reallocations through a general-equilibrium framework in the tradition of Matsuyama (1992). That framework spans open and closed food markets, and the orientation of the estimated signs—warmer nights holding labor on the farm, warmer days pushing it off—places India's districts on the open side, where a closed subsistence economy would reverse both. It guides the signs rather than supplying a structure to estimate; its content is reduced-form panel and long-difference regressions that test those statics.

The approach has limits. The two margins are correlated at 0.93, and collinearity might seem to manufacture their opposing signs; it does not—each already carries its sign, significantly, when it enters alone, so the correlation widens the standard errors without biasing the estimates, though their separation rests on limited residual variation. The divergence further assumes labor and land are gross substitutes ($\sigma > 1$), a keystone the paper does not estimate; the framework guides the signs but is not itself tested; and the uniform town drag softens under a humidity-adjusted wet-bulb correction. Nonlinearity and spatial sorting, held fixed by the linear framework, are left for later work.

The framework's distinctive prediction—that a closed subsistence economy would invert the pattern—is the natural next test: India's pre-1991 Censuses, before liberalization opened the agricultural economy, offer a closed-regime counterfactual to falsify the open-regime reading. The pattern itself is consistent: warmer nights hold labor on the farm without the price and wage distress that warmer days bring, while daytime heat restrains the non-agricultural transition far more strongly. The urban shift into informal services it leaves is less an engine of growth than a physiologically constrained churn for survival.

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Appendix A. Theoretical Framework

Full derivations and proofs are in the supplemental online appendix; this section gives the structure and the predictions it delivers.

Primitives. A representative district produces an agricultural good (relative price p) and a non-agricultural numeraire. Agricultural output combines effective labor H_a and effective land BZ_c in a CES aggregator scaled by total factor productivity,

$$Y_a = A\big[\,\theta\, H_a^{\rho} + (1-\theta)(BZ_c)^{\rho}\,\big]^{1/\rho}, \qquad \rho \equiv \frac{\sigma-1}{\sigma}, \quad \theta \in (0,1), \tag{A1}$$

with $\sigma > 0$ the elasticity of substitution between labor and land, Z_c cultivated area, $A = A(T_{max})$ Hicks-neutral, and $B = B(T_{min})$ land-augmenting. Non-agriculture produces the numeraire under decreasing returns to labor (curvature ν) with capacity A_n, and food carries subsistence content through a demand multiplier $\mathcal{M} > 1$.

Labor identity. The workforce $\bar{L}$ divides across four states whose shares sum to one,

$$l_a + l_c + l_n + l_s = 1, \tag{A2}$$

wage labor l_a, self-cultivation l_c, non-agriculture l_n, and a casual seasonal margin l_s that absorbs the residual. Workers take a wage job when the market wage clears their reservation value, so market supply is $L_a + L_n = \bar{L}\,G(w)$ with elasticity η_s, and mobility equalizes w across sectors. The worker shares in Table A1 are the components of (A2), so they move as one offsetting system.

Closures and shocks. A single openness parameter closes the price,

$$\hat{p} = \chi\,(\hat{Y}_n - \mathcal{M}\,\hat{Y}_a), \qquad \chi \in [0,1], \tag{A3}$$

with $\chi = 0$ the open district ($\hat{p} = 0$, a within-state price taker) and $\chi = 1$ closed. A hat is a log deviation; warming enters as the land-bias shock $b \equiv (B'/B)\,dT_{min}$, the neutral shock $a \equiv (A'/A)\, dT_{max} \leq 0$, and the capacity shock $a_n \equiv \hat{A}_n$. Let $s_Z = 1 - s_L$ be land's revenue share and $\kappa \equiv \sigma / s_Z$.

Micro-to-macro and labor demand. The district responses aggregate micro choices. The farm chooses H_a and cultivated area Z_c to maximize profit $pY_a - wH_a - c(Z_c)$ at given (p, w); its labor first-order condition $p\, \partial Y_a / \partial H_a = w$ yields the labor demand

$$\hat{H}_a = b + \kappa(\hat{p} + a - \hat{w}), \tag{A4}$$

where the night shock b enters directly through factor bias and the day shock a only through the scale term κ. Labor-market clearing pins w and the price closure pins p, mapping the primitives into the district reduced form (A5).

Assumptions. *(A1) Gross substitutes:* $\sigma > 1$. *(A2) Channel assignment:* T_{min} augments land (B'), T_{max} is a Hicks-neutral loss ($A' < 0$), and informal non-agricultural capacity falls under both margins ($a_n < 0$); the data resolve $B' > 0$. Assumption (A1) is the keystone: at $\sigma = 1$ the night shock is observationally neutral and the margins cannot diverge (Lemma A1).

Equilibrium and predictions. In the open district ($\hat{p} = 0$), demand (A4) and market clearing give

$$\hat{L}_a = \frac{\Delta_o'}{\Delta_o}(\kappa a + b), \qquad 0 < \frac{\Delta_o'}{\Delta_o} < 1, \tag{A5}$$

with $\Delta_o, \Delta_o' > 0$ collecting the elasticities, so the two margins carry opposite signs. The framework is built to reproduce five facts: agricultural wage labor rises with night and falls with day warming (panel, long difference, and wet-bulb); the seasonal margin moves oppositely; the harvest price rises under day warming alone; district non-agriculture is flat; and town non-agriculture falls under both margins over long differences. The lemma and propositions below state the comparative statics tested in Section 3.

LEMMA A1 (Cobb–Douglas is silent). *If $\sigma = 1$, the land-bias shock b is observationally equivalent to a neutral productivity shock, and agricultural labor cannot respond with opposite signs to the two margins.*

PROPOSITION A1 (Open-economy divergence). *Under Assumptions (A1)–(A2) with $\chi < \chi^*$, agricultural labor rises with nighttime and falls with daytime warming, $\partial \hat{L}_a / \partial T_{min} > 0$ and $\partial \hat{L}_a / \partial T_{max} < 0$. The seasonal margin moves oppositely, the field wage shares the sign of $\kappa a + b$, and non-agricultural labor moves with $-\hat{w}$.*

PROPOSITION A2 (Openness threshold). *The orientation in Proposition A1 holds if and only if $\chi < \chi^* \equiv 1/(\sigma\mathcal{M})$. Because $\sigma > 1$, the nighttime sign binds first: the night response pins down how open the economy must be.*

PROPOSITION A3 (Closed economy reverses). *In the closed economy ($\chi = 1$) with subsistence ($\mathcal{M} > 1$) and $\sigma > 1$, both signs invert—nighttime warming lowers and daytime warming raises agricultural labor—through the subsistence income effect. The observed orientation therefore places the district in the open regime.*

PROPOSITION A4 (Amplitude). *Rural and urban populations share the signs of Proposition A1; the magnitude scales with the agricultural employment share and the seasonal-supply elasticity η_s, not with the degree of closure. The rural–urban contrast is one of amplitude, not of sign.*

PROPOSITION A5 (Uniform town drag). *Where agriculture is too small to register the reallocation, $a_n < 0$ delivers a uniform contraction of non-agricultural employment under both margins, $\partial\hat{L}_n/\partial T_{min} < 0$ and $\partial\hat{L}_n/\partial T_{max} < 0$, with no day/night divergence.*

TABLE A1. Model predictions: comparative statics

Outcome	Parametric solution	Sign (T_{min}, T_{max})
Cultivated area	extensive margin, both shifters	$(-, -)$
Yield index	$a + s_L(\hat{L}_a - \hat{Z}_c) + s_Z b$	$(\approx 0/+, \approx 0/-)$
Production	$\hat{Y}_a = a + s_L\hat{L}_a + s_Z b$	$(-, -)$
Field wage	$\hat{w} = \phi_a(\kappa a + b)/\Delta_o$	$(\approx 0, -)$
Harvest price	$\hat{p} = \chi(\hat{Y}_n - \mathcal{M}\hat{Y}_a)$	$(\approx 0, +)$
Ag. wage labor	$\hat{L}_a = (\Delta'_o/\Delta_o)(\kappa a + b)$	$(+, -)$
Self-cultivation	night-bias residual	$(-, \approx 0)$
Seasonal	$\propto -\hat{w}$, opposes $\hat{L}_a$	$(-, +)$
District non-agriculture	$\hat{L}_n = -\hat{w}/(1-\nu)$, a_n=0	$(\approx 0, \approx 0)$
Total workers	participation $\propto \hat{w}$	$(+, -)$
Town non-agriculture	$\hat{L}_n = (a_n - \hat{w})/(1-\nu)$, a_n<0	$(-, -)$
Closed-economy counterfactual	χ=1 in the general solution	$(-, +)$ reversed

Notes: Signs are the model's predicted responses of each outcome to nighttime (T_{min}) and daytime (T_{max}) warming, holding the other margin fixed; ≈ 0 marks a response the model leaves weak or indeterminate. $\kappa = \sigma/s_Z$; s_L, s_Z are the labor and land cost shares; ϕ_a is agriculture's weight in labor demand; $\hat{Z}_c$ is the change in cultivated land; and $\Delta_o, \Delta'_o > 0$ collect the supply and demand elasticities. The closed-economy row is the counterfactual the data reject (Proposition A3).

Appendix B. Summary Statistics

Appendix C. Robustness of the District Decomposition

This section reproduces the district decomposition of Table 1 under alternative climate measures, an alternative estimator, and finer geographic and sectoral cuts.

TABLE A2. Town Summary Statistics

	1981	1991	2001	2011
Panel A: All Towns				
Town Characteristics				
Population	42,322.64	65,209.97	74,454.70	71,033.23
Total Workers	12,786.26	19,651.29	23,853.15	25,122.46
Area (sq. km)	14.54	16.18	17.13	14.01
Share of Workers by Category				
Non-Agricultural	0.73	0.74	0.74	0.72
Agricultural Labor	0.12	0.12	0.07	0.08
Cultivation	0.11	0.10	0.06	0.05
Seasonal	0.04	0.04	0.13	0.16
Climatic Indicators (Growing Seasons, 10y MA)				
Degree Days (DDs)	16.00	16.16	16.26	16.53
Daily Temp (° C)	24.07	24.23	24.32	24.58
Nighttime: T_{min} (° C)	18.28	18.46	18.69	19.01
Daytime: T_{max} (° C)	29.85	30.00	29.95	30.15
Daily Precipitation (mm.)	0.98	1.00	0.98	1.01
Observations	3,714	4,402	4,857	7,592
Panel B: Balanced Panel (1981–2011)				
Town Characteristics				
Population	45,259.05	82,584.48	105560.07	131999.55
Total Workers	13,637.92	24,691.57	33,472.11	46,110.79
Area (sq. km)	15.05	18.60	20.55	20.25
Share of Workers by Category				
Non-Agricultural	0.74	0.74	0.76	0.74
Agricultural Labor	0.11	0.12	0.06	0.07
Cultivation	0.12	0.10	0.05	0.05
Seasonal	0.04	0.03	0.12	0.15
Climatic Indicators (Growing Seasons, 10y MA)				
Degree Days (DDs)	15.82	15.96	15.98	16.20
Daily Temp (° C)	23.89	24.03	24.05	24.27
Nighttime: T_{min} (° C)	18.02	18.15	18.28	18.47
Daytime: T_{max} (° C)	29.76	29.92	29.83	30.07
Daily Precipitation (mm.)	0.98	0.97	0.96	0.95
Observations	2,969	2,969	2,969	2,969

Notes: Decadal means for the town panel, computed on the same data the town regressions use (digitized District Census Handbooks, 1981–2011, linked to IMD climate). Panel A covers all towns observed in each Census year (the panel sample); Panel B restricts to the balanced panel of 2,969 towns present in every decade (the long-difference sample). Worker shares are shares of total town workers. Climatic indicators are 10-year moving averages over the growing season—the regressors entering the estimates.

TABLE A3. District Summary Statistics

	1981	1991	2001	2011
Panel A: Rural				
Characteristics				
Population (in Thousands)	1,459.35	1,781.56	1,554.22	1,662.11
Total Workers (in Thousands)	574.58	725.07	662.23	713.37
Share of Workers by Category				
Non-Agricultural	0.19	0.20	0.24	0.23
Agricultural Labor	0.23	0.25	0.17	0.21
Cultivation	0.47	0.44	0.34	0.28
Seasonal	0.11	0.12	0.26	0.28
Climatic Indicators (Growing Seasons, 10y MA)				
Degree Days (DDs)	15.60	15.80	15.74	15.97
Daily Temp (° C)	23.66	23.87	23.80	24.03
Nighttime: T_{min} (° C)	17.73	17.95	17.97	18.15
Daytime: T_{max} (° C)	29.59	29.78	29.63	29.91
Daily Precipitation (mm.)	0.97	0.98	0.97	0.95
Observations	291	293	297	296
Panel B: Urban				
Characteristics				
Population (in Thousands)	462.83	622.66	631.67	812.01
Total Workers (in Thousands)	138.03	187.63	203.32	287.46
Town Counts				
Census Towns	7.58	8.93	10.43	17.51
Statutory Towns	5.33	6.24	4.90	5.72
Share of Workers by Category				
Non-Agricultural	0.81	0.81	0.81	0.77
Agricultural Labor	0.07	0.08	0.04	0.05
Cultivation	0.08	0.07	0.04	0.04
Seasonal	0.03	0.03	0.11	0.14
Climatic Indicators (Growing Seasons, 10y MA)				
Degree Days (DDs)	15.66	15.86	15.80	16.03
Daily Temp (° C)	23.72	23.93	23.87	24.10
Nighttime: T_{min} (° C)	17.80	18.02	18.04	18.22
Daytime: T_{max} (° C)	29.65	29.84	29.69	29.97
Daily Precipitation (mm.)	0.96	0.98	0.97	0.95
Observations	289	293	297	297

Notes: Decadal means for the district panel, computed on the same data the district decomposition uses (digitized District Census Handbooks, 1981–2011, linked to IMD climate), split into the rural (Panel A) and urban (Panel B) sectors the decomposition estimates separately. Worker shares are shares of total sector workers. Climatic indicators are 10-year moving averages over the growing season—the regressors entering the estimates.

TABLE A4. Degree-Days Specification

Panel A. ICRISAT Districts (1981-2011)

	Log Levels (All Grains)			
	Cultivated Area	Yield Index	Field Wage	Harvest Price Index
	(1)	(2)	(3)	(4)
Δ Degree Days	-0.427***	0.014	-0.108	0.064
	(0.087)	(0.066)	(0.091)	(0.048)
Observations	1,243	1,211	1,115	1,191
Dep Var. Mean	5.45	7.14	2.86	5.66

Panel B. Census Districts (1981-2011)

	Share of Workers			
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture
	(1)	(2)	(3)	(4)
Δ Degree Days	-0.020	-0.041**	0.036***	0.025
	(0.018)	(0.018)	(0.014)	(0.018)
Observations	1,187	1,187	1,187	1,187
Dep Var. Mean	0.17	0.32	0.18	0.32
Rainfall Control	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y

Notes: Replicates Table 1 replacing the separate T_{min} and T_{max} terms with a single growing-season degree-days measure (*dds_gs*), averaged over the preceding decade. Decadal panel (1981–2011) with state-year fixed effects; all specifications control for precipitation over the same window. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A5. Wet-Bulb Temperatures

Panel A. ICRISAT Districts (1981-2011)				
	Log Levels (All Grains)			
	Cultivated Area	Yield Index	Field Wage	Harvest Price Index
	(1)	(2)	(3)	(4)
$\Delta\ T^{w}_{min}$	0.090	0.153**	0.137	-0.237***
	(0.078)	(0.077)	(0.124)	(0.060)
$\Delta\ T^{w}_{max}$	-0.579***	-0.194	-0.328*	0.366***
	(0.153)	(0.128)	(0.193)	(0.094)
Observations	1,243	1,211	1,115	1,191
Dep Var. Mean	5.45	7.14	2.86	5.66
F-stat ($T^{w}_{min} = T^{w}_{max}$)	9.01	3.07	2.28	16.59
Panel B. Census Districts (1981-2011)				
	Share of Workers			
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture
	(1)	(2)	(3)	(4)
$\Delta\ T^{w}_{min}$	-0.083***	-0.013	0.086***	0.011
	(0.022)	(0.024)	(0.017)	(0.022)
$\Delta\ T^{w}_{max}$	0.095***	-0.028	-0.079***	0.009
	(0.035)	(0.035)	(0.025)	(0.036)
Observations	1,187	1,187	1,187	1,187
Dep Var. Mean	0.17	0.32	0.18	0.32
F-stat ($T^{w}_{min} = T^{w}_{max}$)	10.68	0.08	16.27	0.00
Rainfall Control	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y

Notes: Replicates Table 1 using growing-season wet-bulb minimum and maximum temperatures (T^{w}_{min}, T^{w}_{max}) in place of the surface-air measures, averaged over the preceding decade. Decadal panel (1981–2011) with state-year fixed effects and precipitation controls. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A6. Wet-Bulb Temperatures: Long-Difference Labor Reallocation

	Δ Share of Workers					
	Seasonal		Ag. Labor		Non-Agriculture	
	District	Town	District	Town	District	Town
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\ T^{w}_{min}$	-0.147***	-0.042**	0.159***	0.003	0.001	0.087***
	(0.044)	(0.021)	(0.033)	(0.023)	(0.047)	(0.031)
$\Delta\ T^{w}_{max}$	0.144*	0.051	-0.165**	0.119***	0.049	-0.249***
	(0.086)	(0.039)	(0.065)	(0.042)	(0.097)	(0.061)
Observations	293	2,969	293	2,969	293	2,969
Dep Var. Mean	0.14	0.10	-0.05	-0.05	0.05	0.01
F-stat ($T^{w}_{min} = T^{w}_{max}$)	5.42	2.63	11.95	3.47	0.12	14.29
Rainfall Control	Y	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y	Y

Notes: Replicates Table 2 (single long difference, 1981–2011, state fixed effects) using growing-season wet-bulb minimum and maximum temperatures in place of the surface-air measures. Under humidity adjustment the town non-agricultural share diverges—warmer wet-bulb nights *raise* it and hotter wet-bulb days lower it—in contrast to the uniform drag under dry-bulb temperatures in Table 2; see Section 3. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A7. Long-Difference Estimates Within Districts

Panel A. ICRISAT Districts (Long Difference, 1981-2011)				
	Δ Log Levels (All Grains)			
	Cultivated Area	Yield Index	Field Wage	Harvest Price Index
	(1)	(2)	(3)	(4)
$\Delta\ T_{min}$	-0.755***	0.084	-0.235*	-0.050
	(0.192)	(0.159)	(0.121)	(0.063)
$\Delta\ T_{max}$	-0.844***	-0.200	-0.267*	0.094
	(0.264)	(0.225)	(0.153)	(0.091)
Observations	311	303	259	294
Dep Var. Mean	-0.08	0.37	1.92	1.36
F-stat ($T_{min} = T_{max}$)	0.42	4.70	0.03	2.21
Panel B. Census Districts (Long Difference, 1981-2011)				
	Δ Share of Workers			
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture
	(1)	(2)	(3)	(4)
$\Delta\ T_{min}$	-0.074***	-0.024	0.075***	0.026
	(0.027)	(0.031)	(0.021)	(0.028)
$\Delta\ T_{max}$	0.049	-0.008	-0.062**	0.020
	(0.033)	(0.038)	(0.025)	(0.037)
Observations	293	293	293	293
Dep Var. Mean	0.14	-0.14	-0.05	0.05
F-stat ($T_{min} = T_{max}$)	12.34	0.16	26.07	0.03
Rainfall Control	Y	Y	Y	Y
State FE	Y	Y	Y	Y

Notes: Replicates Table 1 using a single long difference (1981–2011) of each outcome on the long-run change in T_{min} and T_{max}, with state fixed effects and precipitation controls, in place of the decadal panel. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A8. District Worker Shares, Long Difference (Rural vs Urban)

Panel A. Rural Workers (long difference, 1981–2011)

	Δ Share of Workers				
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
Δ T_{min}	-0.064**	-0.035	0.096***	0.007	0.211
	(0.029)	(0.031)	(0.023)	(0.020)	(0.231)
Δ T_{max}	0.056	0.011	-0.068**	-0.001	0.025
	(0.036)	(0.041)	(0.027)	(0.029)	(0.259)
Observations	291	291	291	291	293
Dep Var. Mean	0.15	-0.15	-0.05	0.04	0.06
F-stat ($T_{min} = T_{max}$)	11.36	1.24	30.87	0.07	0.48

Panel B. Urban Workers (long difference, 1981–2011)

	Δ Share of Workers				
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
Δ T_{min}	-0.028**	-0.013	0.034***	0.007	0.632*
	(0.013)	(0.014)	(0.013)	(0.027)	(0.361)
Δ T_{max}	0.006	0.015	0.006	-0.028	0.305
	(0.016)	(0.016)	(0.019)	(0.038)	(0.383)
Observations	291	291	291	291	293
Dep Var. Mean	0.10	-0.04	-0.03	-0.02	0.31
F-stat ($T_{min} = T_{max}$)	3.23	1.69	1.89	0.69	0.60
Rainfall Control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: Decomposes the district worker-share results of Table 1 (Panel B) by place of residence over a single long difference (1981–2011): Panel A reports the change in shares among rural workers and Panel B among urban workers. The four mutually exclusive Census categories are exhaustive and sum to one, so the share coefficients sum to approximately zero; column 5, set apart by a gap, reports the change in log total workers (the extensive margin). All specifications include state fixed effects and control for the long-run change in precipitation. The F-statistic tests $T_{min} = T_{max}$. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A9. District Log Population (Total, Rural, Urban)

Panel A. Decadal Panel (1981–2011)

	Δ Log Population		
	Total	Rural	Urban
	(1)	(2)	(3)
Δ T_{min}	0.147*	0.132	0.303**
	(0.088)	(0.086)	(0.149)
Δ T_{max}	-0.032	-0.004	0.025
	(0.085)	(0.083)	(0.165)
Observations	1,187	1,187	1,187
F-stat ($T_{min} = T_{max}$)	1.95	1.28	1.67

Panel B. Long Difference (1981–2011)

	Δ Log Population		
	Total	Rural	Urban
	(1)	(2)	(3)
Δ T_{min}	0.236	0.127	0.650*
	(0.245)	(0.234)	(0.361)
Δ T_{max}	0.084	0.080	0.313
	(0.277)	(0.269)	(0.387)
Observations	293	293	293
F-stat ($T_{min} = T_{max}$)	0.28	0.03	0.64
Rainfall Control	Y	Y	Y
State FE / Trend	Y	Y	Y

Notes: Estimates of a 1°C rise in growing-season nighttime (T_{min}) and daytime (T_{max}) temperature on log district population, for the total, rural, and urban populations. Panel A is the decadal panel (1981–2011) with state-year fixed effects; Panel B is the single long difference (1981–2011) with state fixed effects. All specifications control for precipitation. The F-statistic tests $T_{min} = T_{max}$. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A10. Town-Level Worker Shares

Panel A. Census Towns – Decadal Panel (1981-2011)

	Share of Workers			
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture
	(1)	(2)	(3)	(4)
$\Delta\ T_{min}$	0.002	-0.008**	0.017***	-0.012
	(0.006)	(0.004)	(0.005)	(0.008)
$\Delta\ T_{max}$	0.016**	0.004	-0.007	-0.012
	(0.007)	(0.005)	(0.007)	(0.009)
Observations	16,215	16,215	16,215	16,215
Dep Var. Mean	0.10	0.08	0.09	0.73
F-stat ($T_{min} = T_{max}$)	2.58	2.83	7.22	0.00

Panel B. Census Towns – Long Difference (1981-2011)

	Δ Share of Workers			
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture
	(1)	(2)	(3)	(4)
$\Delta\ T_{min}$	-0.016	-0.009	0.063***	-0.038**
	(0.012)	(0.011)	(0.012)	(0.018)
$\Delta\ T_{max}$	0.017	0.022	0.041***	-0.078***
	(0.014)	(0.014)	(0.016)	(0.022)
Observations	2,969	2,969	2,969	2,969
Dep Var. Mean	0.10	-0.06	-0.05	0.01
F-stat ($T_{min} = T_{max}$)	4.06	3.38	1.61	2.78
Rainfall Control	Y	Y	Y	Y
State FE / Trend	Y	Y	Y	Y

Notes: Reproduces the worker-share decomposition at the Census town level. Panel A reports decadal-panel estimates (state-year fixed effects); Panel B reports the long difference (1981–2011, state fixed effects). All specifications control for precipitation. Standard errors clustered by town in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

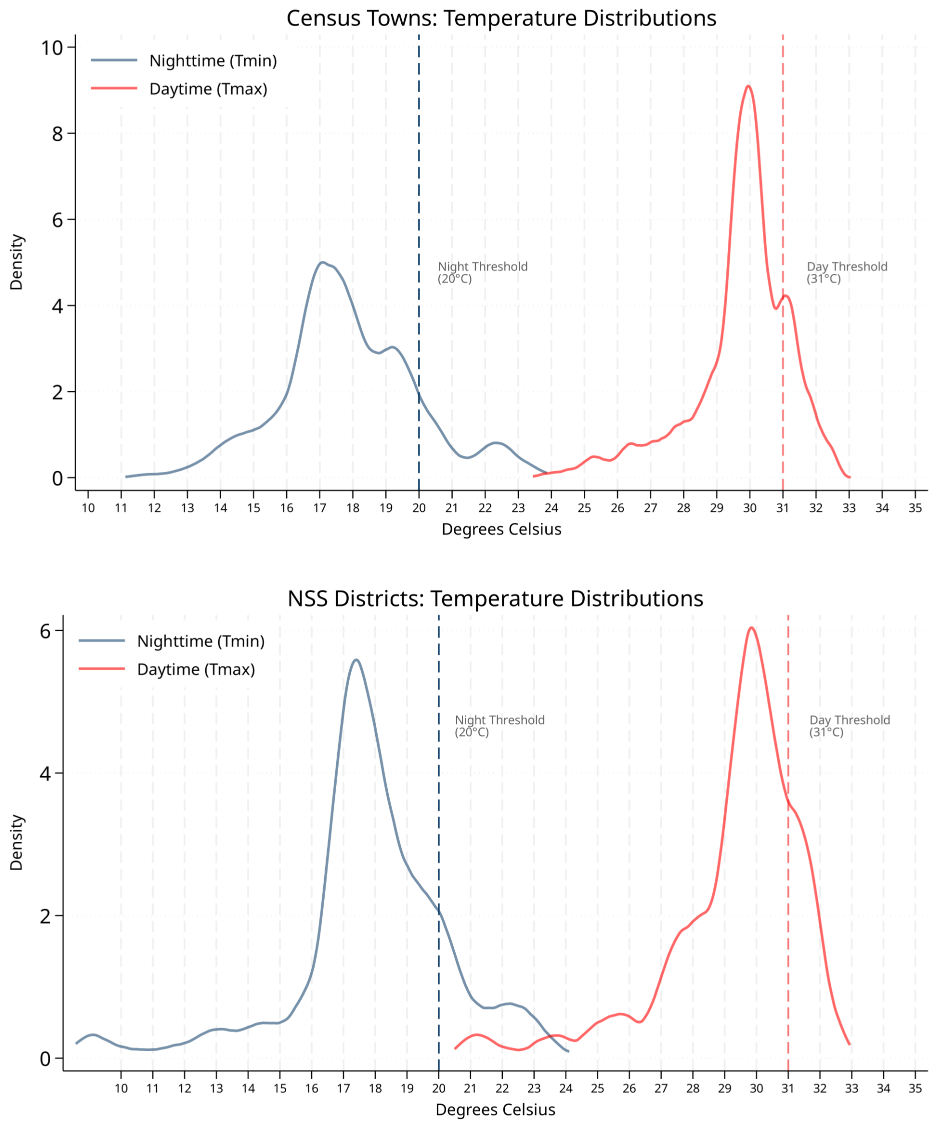
TABLE A11. Urban Worker Shares (NSS/IPUMS), Census-Comparable

	Share of Workers					
			of which non-ag:			
	Cultivation	Ag. Labour	Informal	Formal	Log Ag-Labour Wage	Log Total Workers
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\ T_{min}$	-0.045	-0.010	0.048**	0.007	0.366	-0.024
	(0.027)	(0.024)	(0.023)	(0.016)	(0.305)	(0.059)
$\Delta\ T_{max}$	0.024	-0.043	0.005	0.014	-0.321	0.156
	(0.046)	(0.028)	(0.030)	(0.032)	(0.440)	(0.139)
Observations	1,228	1,228	1,228	1,228	704	1,228
Dep Var. Mean	0.39	0.21	0.21	0.19	5.36	13.08
F-stat ($T_{min} = T_{max}$)	1.99	1.06	2.03	0.04	2.14	1.58
Rainfall Control	Y	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y	Y

Notes: Reproduces the worker-share decomposition for urban households in the NSS Employment–Unemployment rounds (IPUMS-International, 1987–2009), classified to census-comparable categories: cultivators (self-cultivators and unpaid family workers), agricultural wage labour, and non-agriculture (everything else), the last split into informal (construction, mining, and low-skill services) and formal (manufacturing and high-skill services). Decadal panel with state-year fixed effects and precipitation controls; T_{min} and T_{max} are growing-season means over the preceding decade. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix D. Temperature Distributions and Maps

FIGURE A1. Thermal Zones and Physiological Limits

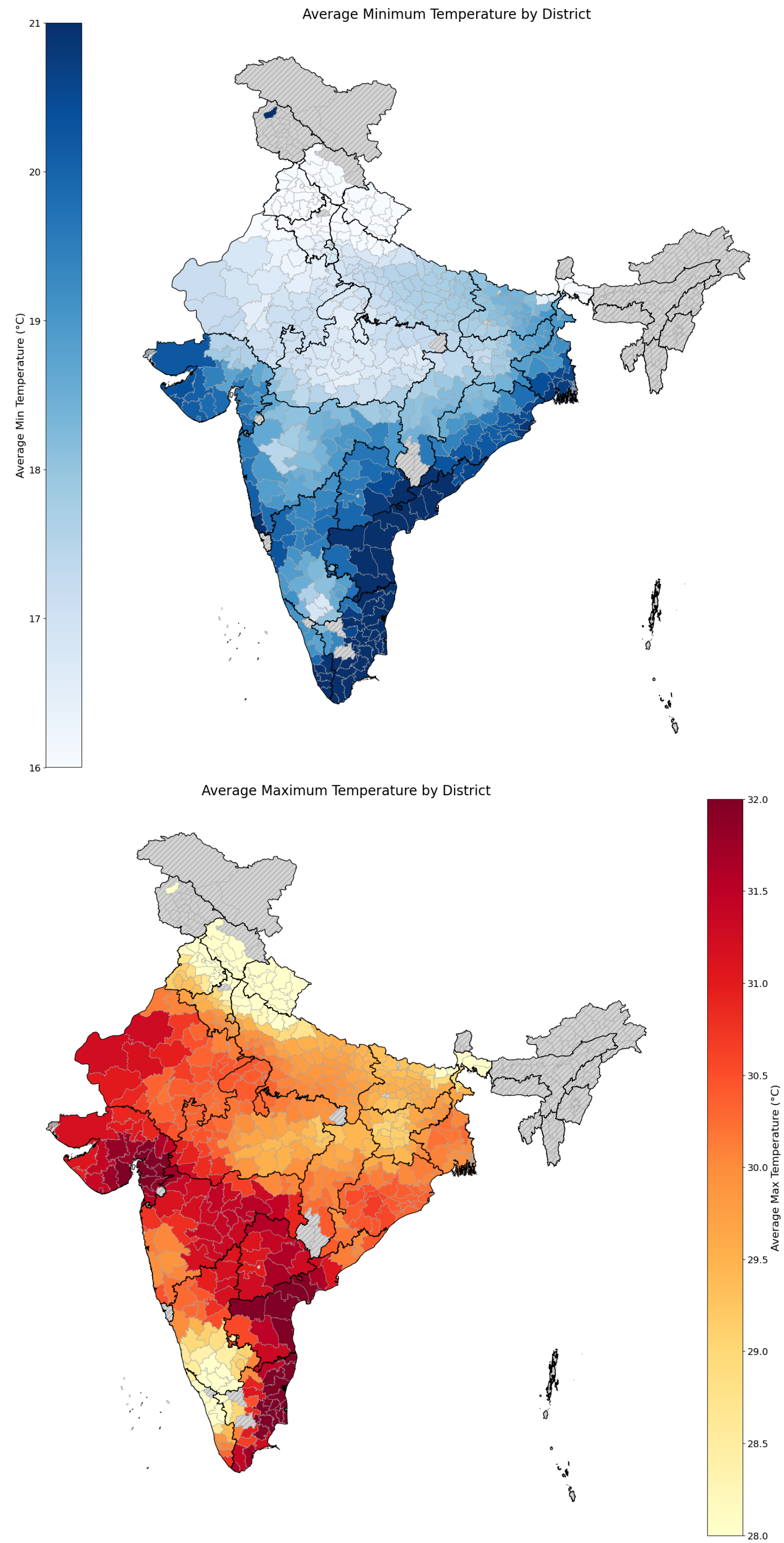


Notes: Kernel densities of long-run surface-air minimum and maximum temperatures across the Census town sample (top) and the NSS/IPUMS urban household sample (bottom), with the night (20°C) and day (31°C) thresholds marked.

Appendix E. Monte Carlo Inference

To choose the panel estimator I ask which standard-error and trend specification has correct *size*: how often it rejects a true null. In each of 1,000 simulations I replace the climate regressor with a placebo built by randomly permuting the real climate series

FIGURE A2. Long-run Temperatures in Districts (1981-2011)



Notes: District-level long-run growing-season surface-air minimum (top) and maximum (bottom) temperatures.

across districts—the T_{min}, T_{max}, and rainfall block is moved together, so the placebo keeps the regressor's decadal persistence, cross-variable correlation, and spatial distribution but is unrelated to the outcome by construction. I regress the real outcome on this placebo and record whether the joint null that both temperature coefficients are zero is rejected at the 5% level; under valid inference the rejection rate should be about 5%. Tables A12–A14 report these rates for the ICRISAT, Census-district, and Census-town outcomes of Table 1, using the baseline surface-air growing-season regressor set. The pattern is consistent across outcomes: because the placebo is persistent, unadjusted and heteroskedasticity-robust errors over-reject by a factor of three or more, and a purely spatial Conley correction does not fix this; clustering by district brings the size close to nominal, and adding state-specific linear trends (or state-by-year fixed effects) keeps it near 5% while district-specific trends over-absorb. This motivates the preferred specification used throughout—district-clustered standard errors with state linear trends. The Census-town panel, in which towns nest within district-years, additionally identifies the district-by-year fixed-effects row that saturates the district-level panel; it too holds size near nominal under the preferred estimator.

TABLE A12. Monte Carlo rejection rates: ICRISAT district outcomes

	Cropped area	Yield	Production	Field wage	Harvest price
No adjustment (homoskedastic)	0.171	0.095	0.170	0.061	0.133
Heteroskedasticity-robust	0.120	0.086	0.122	0.062	0.098
Cluster by district	0.050	0.044	0.055	0.056	0.050
Cluster by district + district linear trends	0.015	0.019	0.018	0.024	0.031
Cluster by district + state linear trends	**0.050**	**0.043**	**0.053**	**0.052**	**0.033**
Cluster by district + state-by-year FE	0.049	0.036	0.053	0.046	0.025
Cluster by district + district-by-year FE	–	–	–	–	–
Cluster by state	0.046	0.077	0.060	0.064	0.074
Conley spatial HAC (50 km)	0.214	0.163	0.197	0.128	0.185
Conley spatial HAC (100 km)	0.215	0.162	0.199	0.130	0.189
Conley spatial HAC (200 km)	0.225	0.179	0.220	0.136	0.218
Conley spatial HAC (50 km, 2 lags)	0.134	0.101	0.144	0.082	0.107
Conley spatial HAC (50 km, 4 lags)	0.053	0.048	0.062	0.057	0.054
Replications			1000		

Notes: Each cell is the share of simulations in which the joint null that both temperature coefficients equal zero is rejected at the 5% level, for the ICRISAT outcomes of Table 1. In each simulation the placebo regressor is built by randomly permuting the real climate series (the T_{min}/T_{max}/rainfall block, moved together) across districts, preserving its 10-year moving-average persistence and spatial distribution while breaking any true link to the outcome; a correctly sized estimator therefore rejects in roughly 5% of simulations. Rows vary the standard-error and trend specification, for the baseline surface-air growing-season regressor set. Conley rows sweep the spatial radius (50–200 km) at zero time lags and, at 50 km, the number of decadal time lags. District-by-year fixed effects saturate the district-level panel and are not estimable (shown as –). The preferred estimator—clustering by district with state-specific linear trends (bold)—holds size near 5%, whereas unadjusted and heteroskedasticity-robust standard errors over-reject.

TABLE A13. Monte Carlo rejection rates: Census district outcomes

	Seasonal	Cultivators	Ag. labour	Non-ag.	Total workers
No adjustment (homoskedastic)	0.172	0.165	0.168	0.169	0.154
Heteroskedasticity-robust	0.122	0.131	0.128	0.129	0.148
Cluster by district	0.056	0.055	0.056	0.060	0.056
Cluster by district + district linear trends	0.021	0.020	0.017	0.013	0.017
Cluster by district + state linear trends	**0.065**	**0.044**	**0.063**	**0.057**	**0.052**
Cluster by district + state-by-year FE	0.054	0.047	0.053	0.047	0.053
Cluster by district + district-by-year FE	–	–	–	–	–
Cluster by state	0.090	0.078	0.089	0.062	0.065
Conley spatial HAC (50 km)	0.210	0.201	0.217	0.221	0.251
Conley spatial HAC (100 km)	0.220	0.212	0.220	0.231	0.264
Conley spatial HAC (200 km)	0.244	0.248	0.259	0.241	0.280
Conley spatial HAC (50 km, 2 lags)	0.119	0.119	0.121	0.118	0.106
Conley spatial HAC (50 km, 4 lags)	0.059	0.059	0.058	0.064	0.061
Replications			1000		

Notes: As in Table A12, for the Census district worker-share and log-employment outcomes of Table 1. Each cell is the rejection rate of the joint null on the two temperature coefficients under a placebo regressor formed by permuting the real climate series across districts. The preferred specification (cluster by district + state linear trends, bold) keeps the empirical size near the nominal 5%.

TABLE A14. Monte Carlo rejection rates: Census town outcomes

	Seasonal	Cultivators	Ag. labour	Non-ag.	Total workers
No adjustment (homoskedastic)	0.088	0.144	0.126	0.121	0.127
Heteroskedasticity-robust	0.073	0.106	0.098	0.093	0.080
Cluster by district	0.040	0.052	0.052	0.054	0.048
Cluster by district + district linear trends	0.045	0.041	0.044	0.058	0.036
Cluster by district + state linear trends	**0.037**	**0.055**	**0.041**	**0.059**	**0.048**
Cluster by district + state-by-year FE	0.039	0.048	0.047	0.064	0.048
Cluster by district + district-by-year FE	0.053	0.046	0.059	0.060	0.049
Cluster by state	0.083	0.093	0.068	0.077	0.073
Conley spatial HAC (50 km)	0.154	0.192	0.178	0.168	0.154
Conley spatial HAC (100 km)	0.150	0.209	0.182	0.180	0.158
Conley spatial HAC (200 km)	–	–	–	–	–
Conley spatial HAC (50 km, 2 lags)	0.074	0.106	0.106	0.099	0.110
Conley spatial HAC (50 km, 4 lags)	0.043	0.049	0.050	0.051	0.049
Replications			1000		

Notes: As in Table A12, for the Census town worker-share and log-employment outcomes of Table 2. The panel unit is the town, and the placebo permutes each town's climate series across towns; standard errors cluster at the district level. Because towns nest within district-years, the district-by-year fixed-effects row is estimable here (unlike the district-level panel). The 200 km Conley row is omitted (–) because the spatial-HAC estimator over the full set of towns is computationally prohibitive. The preferred specification (cluster by district + state linear trends, bold) keeps the empirical size near the nominal 5%.